

Series Worksheet

Do the following series converge or diverge? State the test you used. For convergent series, indicate if they converge conditionally or absolutely.

1.
$$\sum_{k=1}^{\infty} \frac{\cos(3k)}{k^2}$$

2.
$$\sum_{k=1}^{\infty} \frac{k-1}{k^{3/2}}$$

3.
$$\sum_{k=3}^{\infty} \frac{4}{k\sqrt{\ln(k)}}$$

4.
$$\sum_{k=1}^{\infty} \frac{(-1)^k 7^{2k}}{(2k)!}$$

5.
$$\sum_{k=1}^{\infty} \sqrt[k]{k}$$

6.
$$\sum_{k=1}^{\infty} \frac{\ln(k^2)}{k}$$

7. State and explain the Zero-Infinity Comparison Test